

Compound assignment / short hand operators: Here we are using assignment operator with the combination of other operators as follows.

`+=, -=, *=, %/, <<=, >>=,....`

Eg:

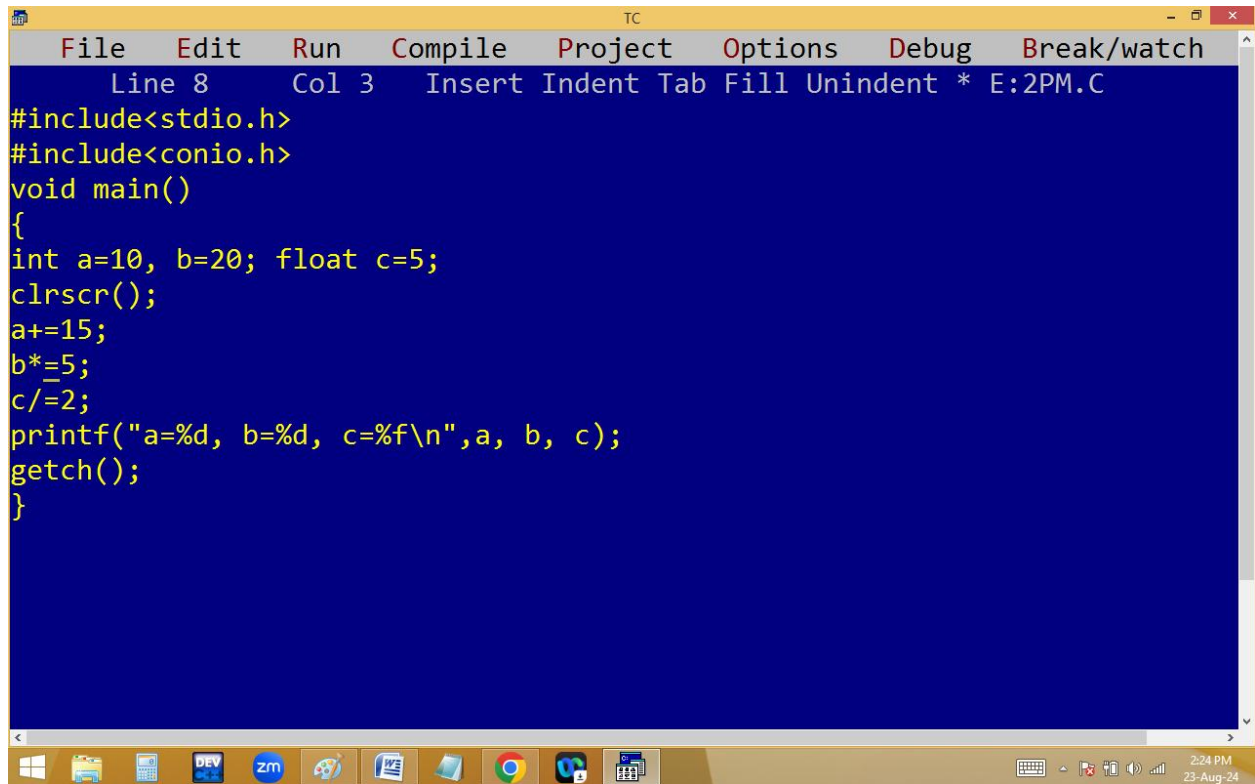
`int a=10, b=20;`

`float c=5;`

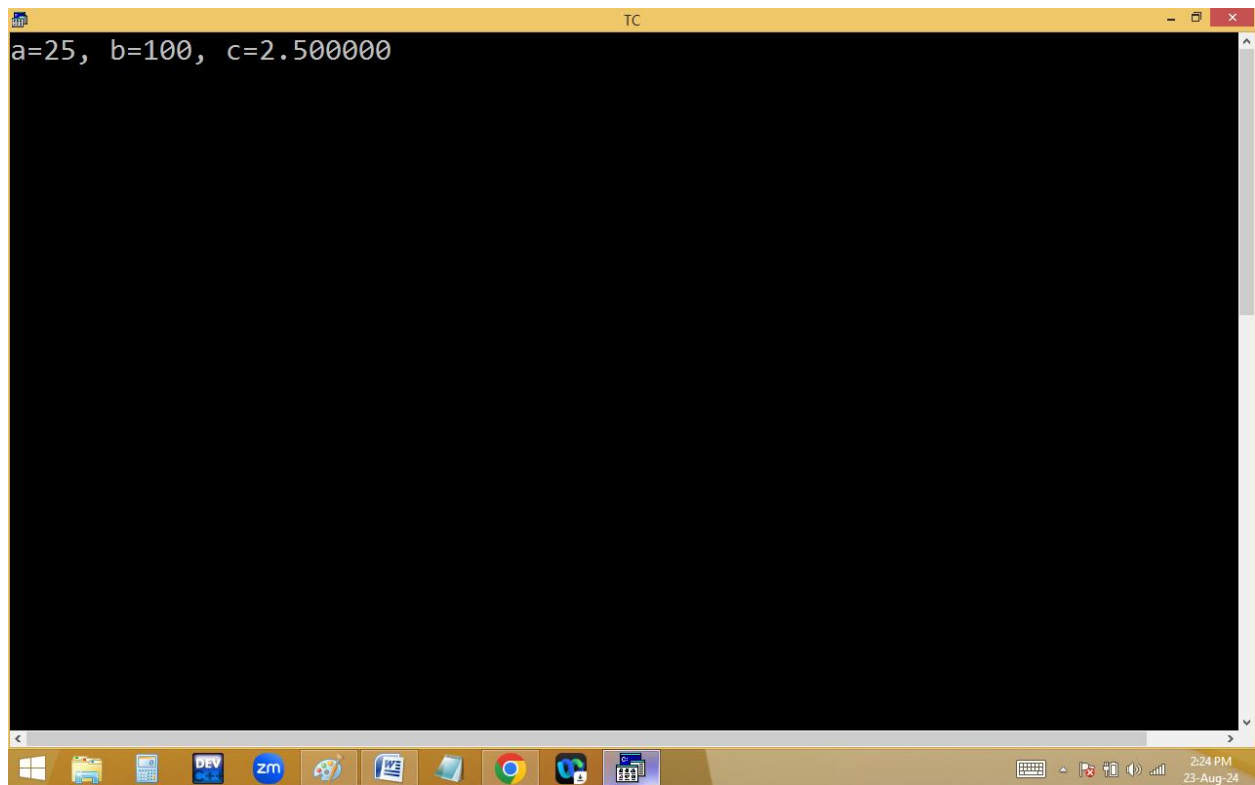
`a+=15; i.e. a=a+15 → a = 10 + 15 → 25`

`b*=5; i.e. b=b*5 → b = 20 * 5 = 100`

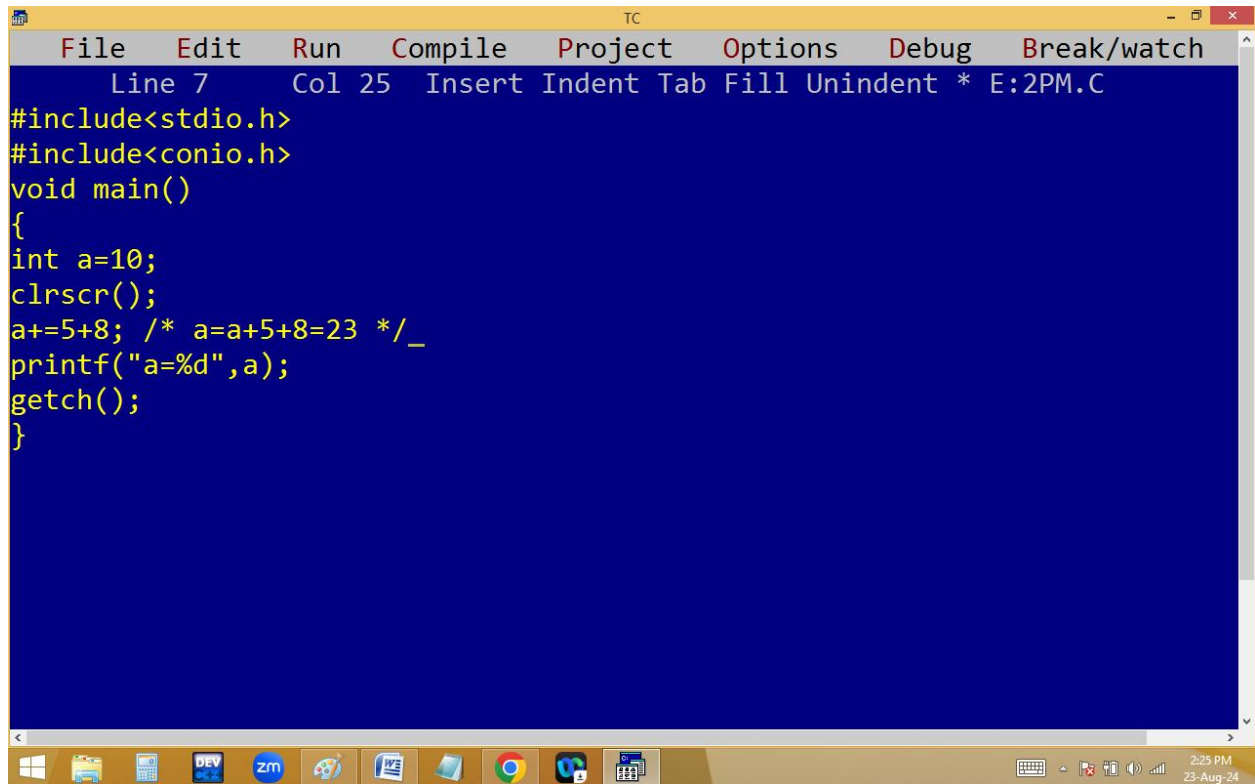
`c/=2; i.e. c=c/2 → c = 5/2 → 2.500000`



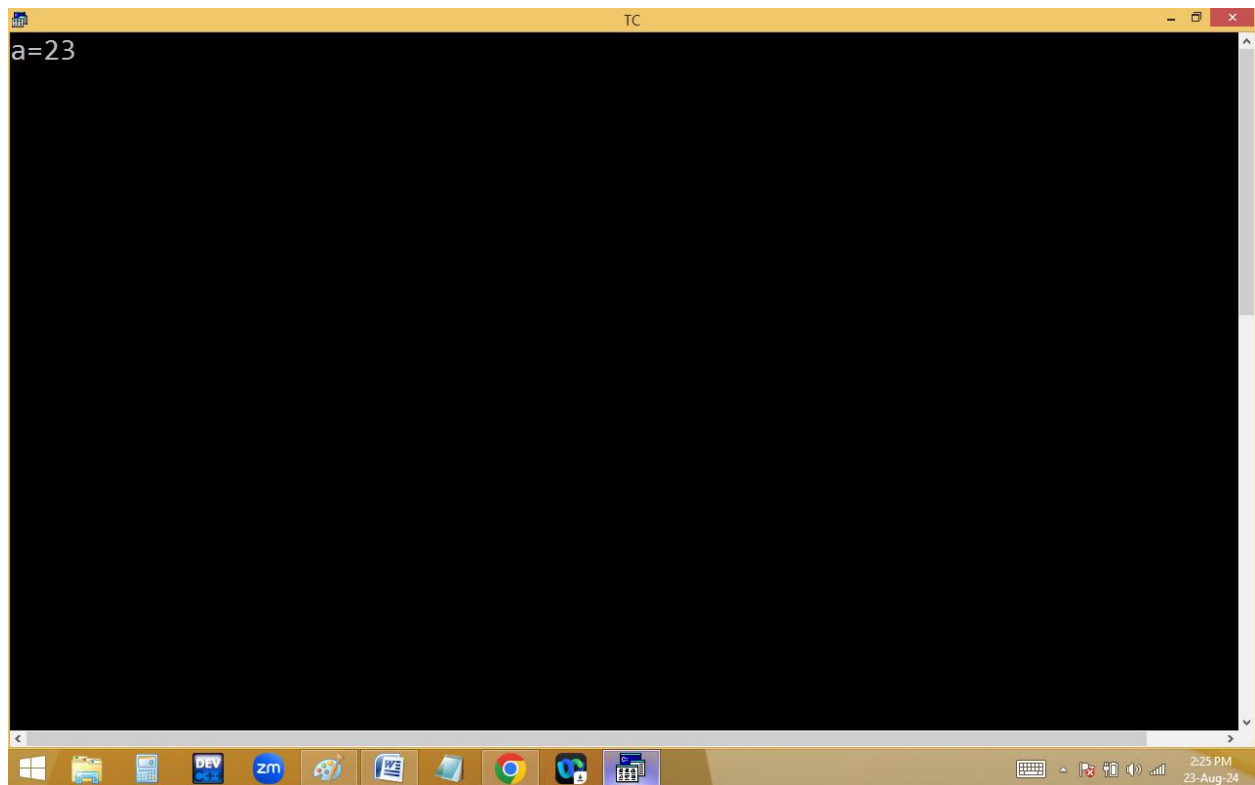
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 3 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=20; float c=5;
clrscr();
a+=15;
b*=5;
c/=2;
printf("a=%d, b=%d, c=%f\n",a, b, c);
getch();
}
```



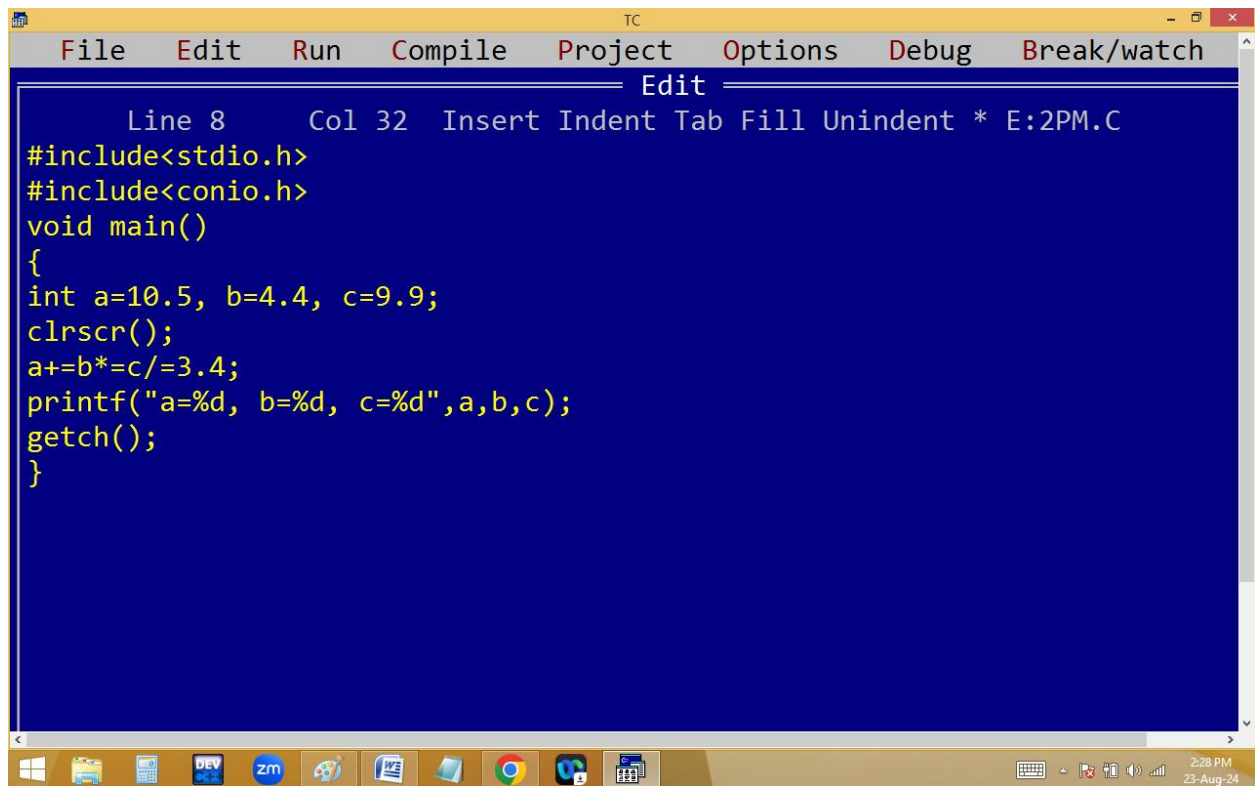
```
TC
a=25, b=100, c=2.500000
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 25 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a+=5+8; /* a=a+5+8=23 */
printf("a=%d",a);
getch();
}
```



```
TC
a=23
```



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The "Edit" menu is open, showing options: "Line 8", "Col 32", "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E:2PM.C". The main editor area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=10.5, b=4.4, c=9.9;
    clrscr();
    a+=b*c/=3.4;
    printf("a=%d, b=%d, c=%d",a,b,c);
    getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, DEV C++, zm, a network icon, a folder icon, Google Chrome, a task manager icon, and a calendar icon. The system clock shows "2:28 PM" and "23-Aug-24".

```
TC
a=18, b=8, c=2_
```

int a=10.~~5~~

b=4.~~4~~

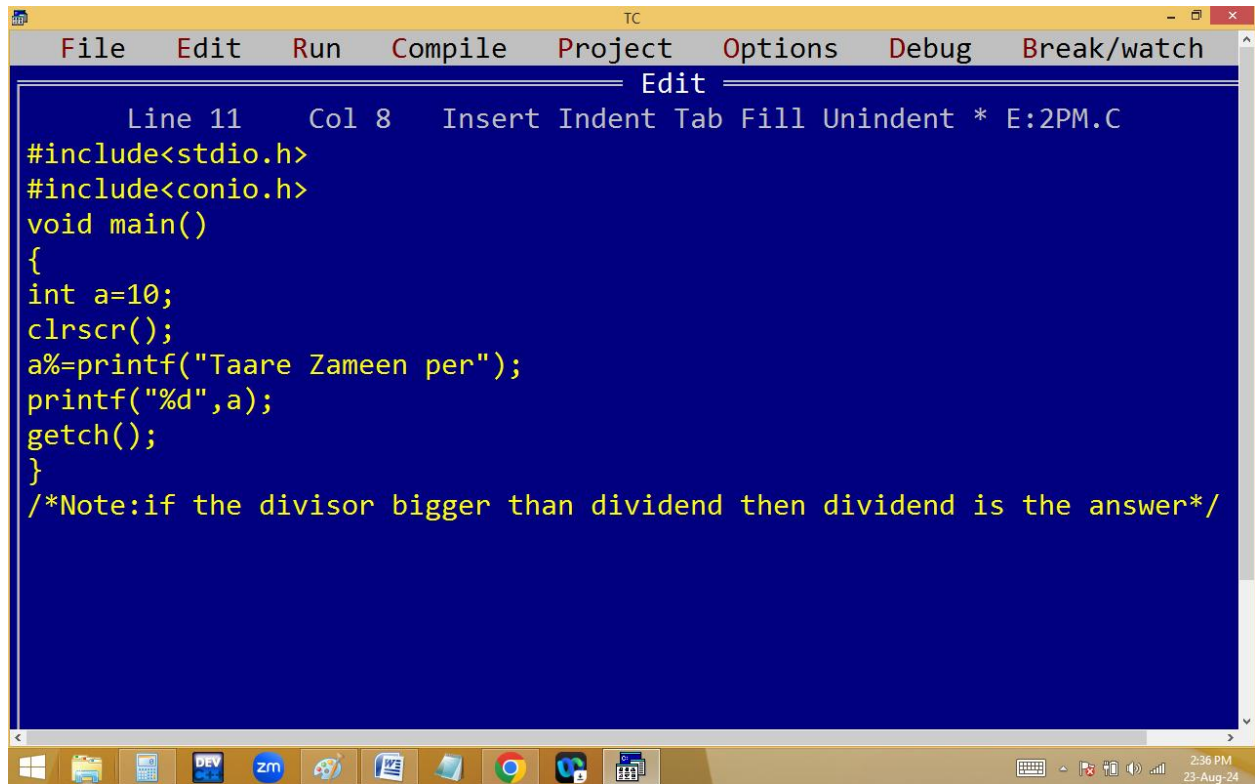
c=9.~~9~~

a+=b*=c/=3.4;

$c = c / 3.4 \implies 9/3.4 \implies 2$ ✓

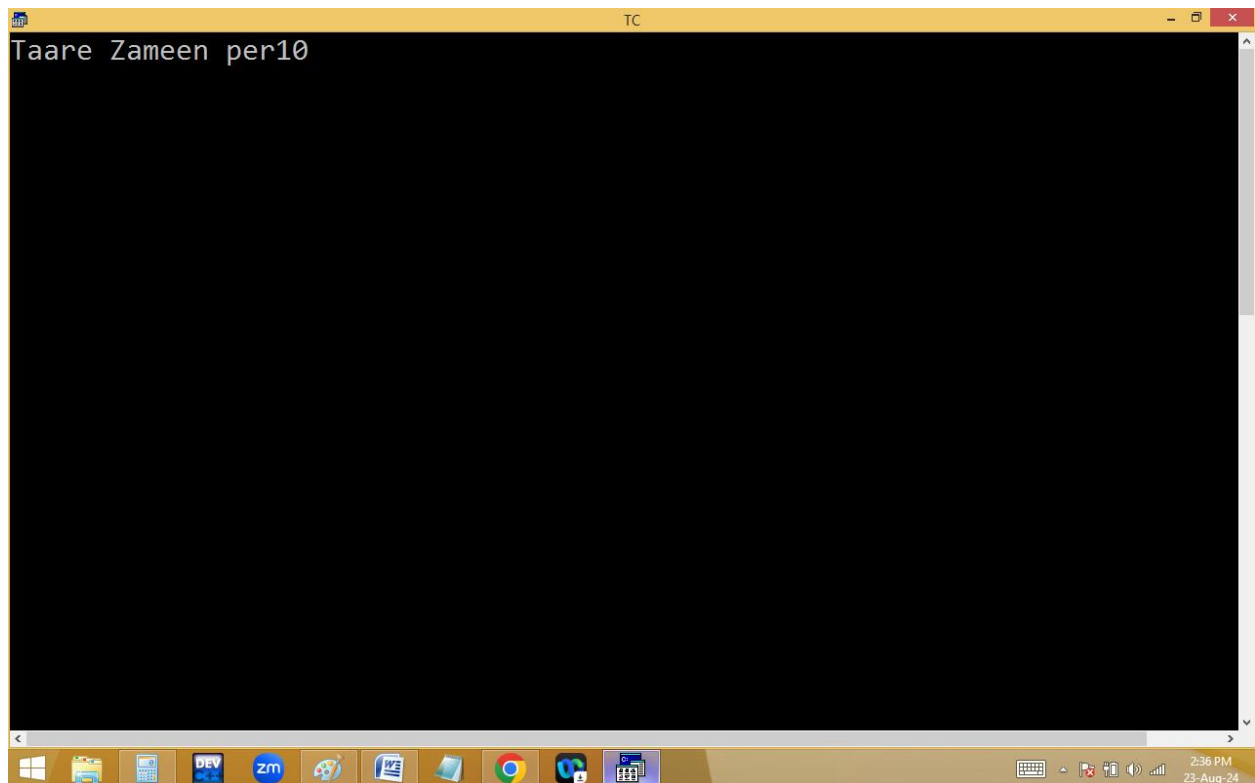
$b * c \implies b = b * c \implies 4 * 2 = 8$ ✓

$a += b \implies a = a + b \implies 10 + 8 = 18$ ✓



The screenshot shows the Turbo C++ IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main window displays a C program in a blue editor. The code includes headers for stdio.h and conio.h, defines a main function, declares an integer variable 'a' with a value of 10, clears the screen, prints the text 'Taare Zameen per', prints the value of 'a', and waits for a key press. A comment at the bottom explains the logic for division based on the relative sizes of the divisor and dividend. The status bar at the bottom shows the time as 2:36 PM on 23-Aug-24.

```
Line 11 Col 8 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=10;
    clrscr();
    a%=printf("Taare Zameen per");
    printf("%d",a);
    getch();
}
/*Note:if the divisor bigger than dividend then dividend is the answer*/
```



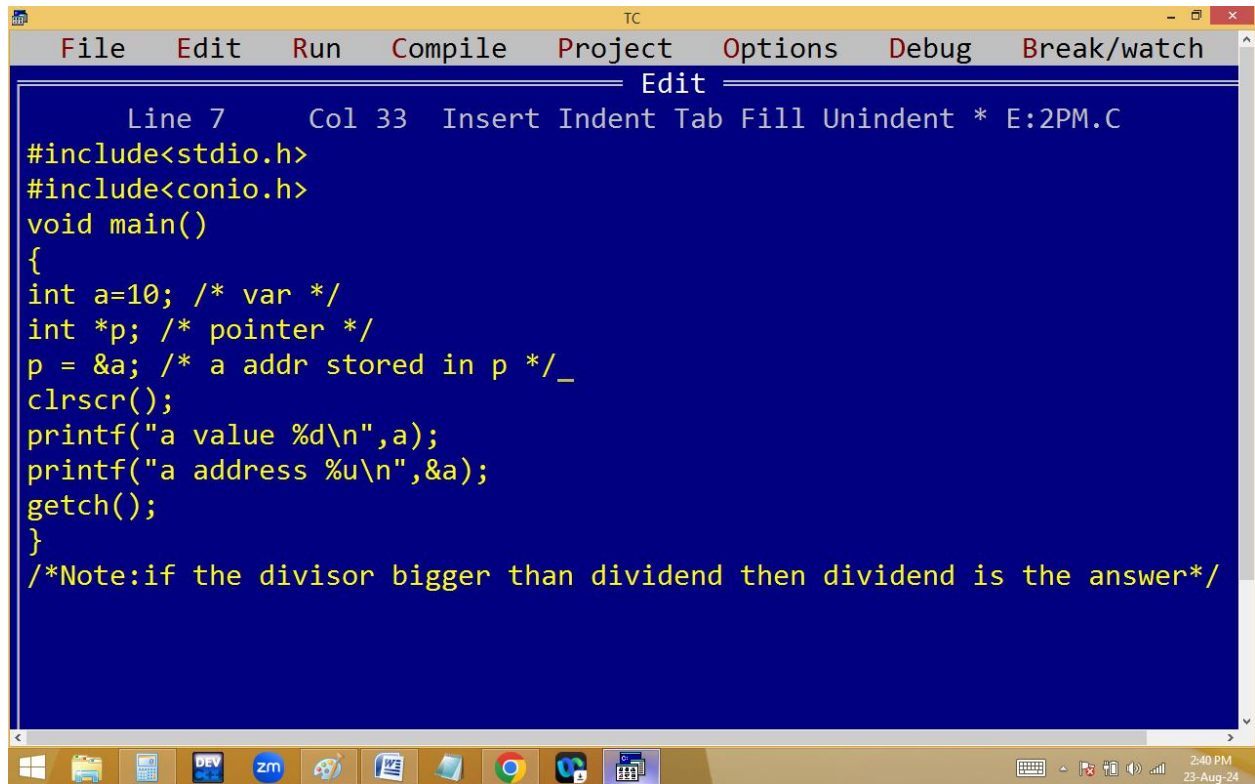
The screenshot shows the Turbo C++ IDE with the same menu bar and toolbar. The main window displays the output of the program, which is the text 'Taare Zameen per10'. The status bar at the bottom shows the time as 2:36 PM on 23-Aug-24.

```
Taare Zameen per10
```

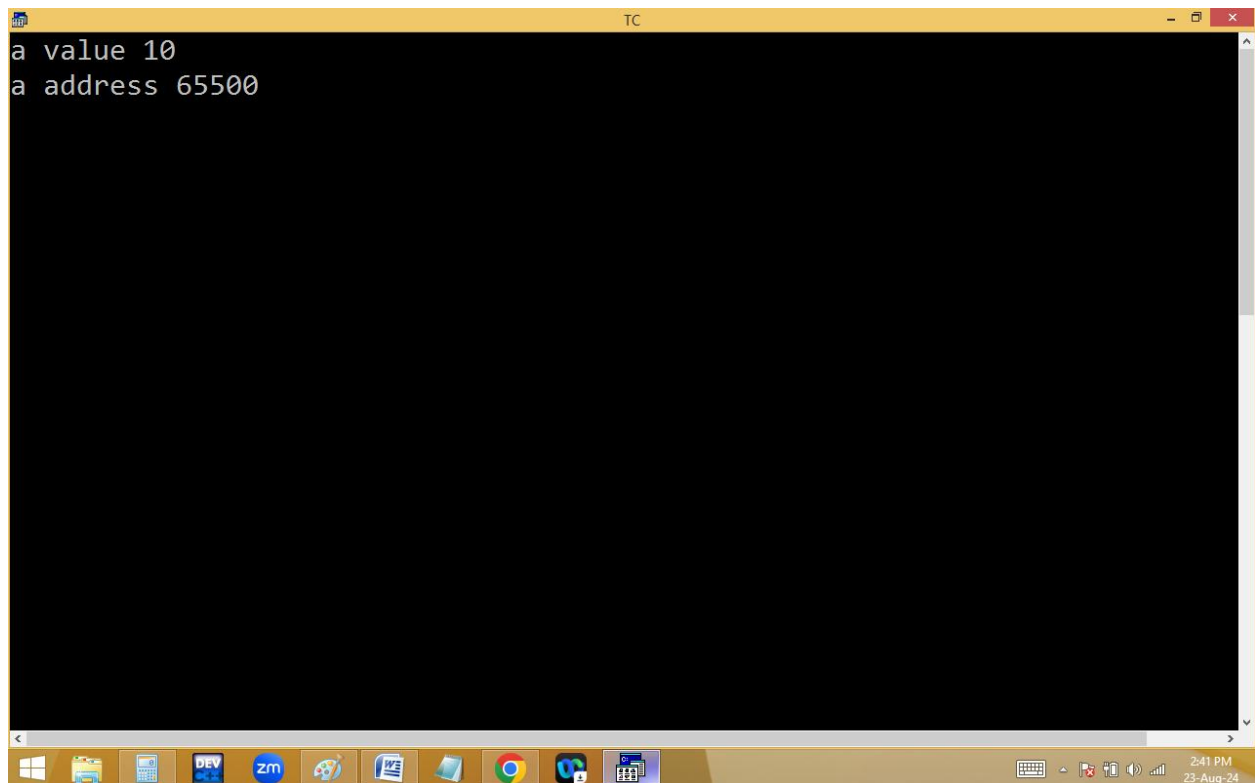
Address operators:

1.& - Address of variable

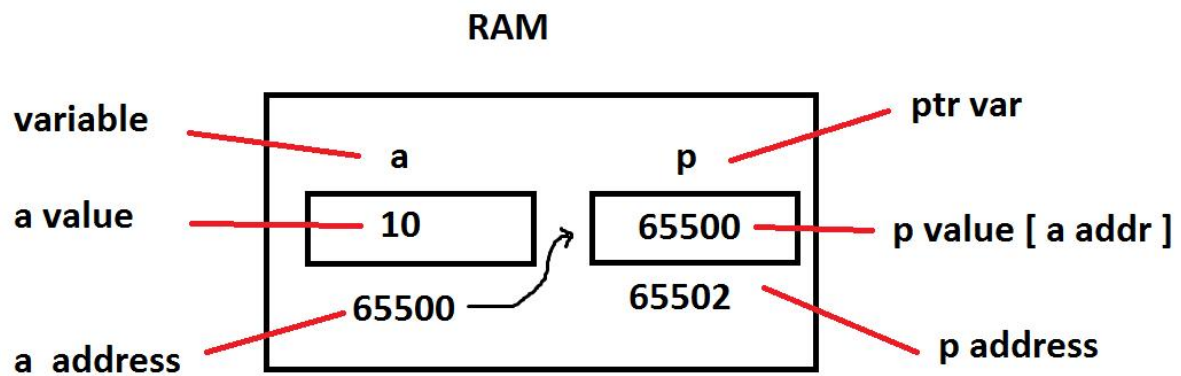
2.* - address of another variable [pointer]



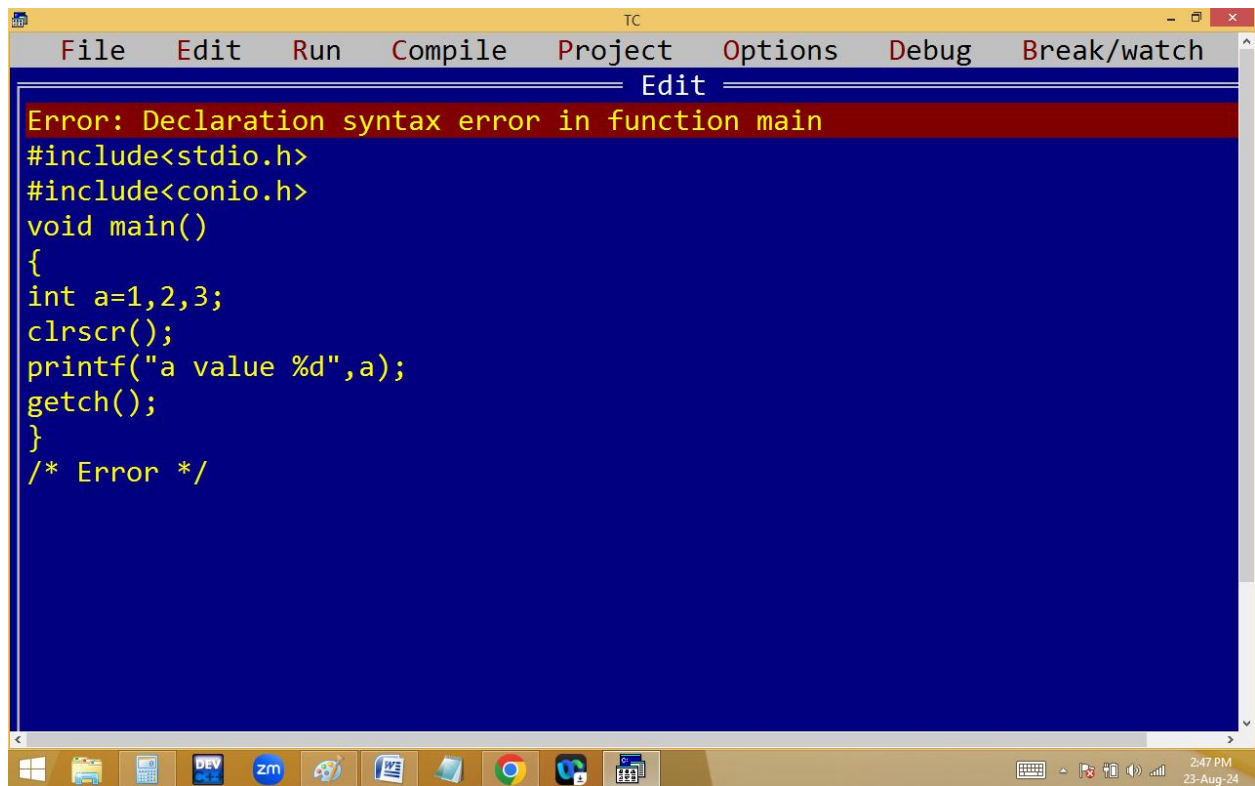
```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 7 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; /* var */
int *p; /* pointer */
p = &a; /* a addr stored in p */_
clrscr();
printf("a value %d\n",a);
printf("a address %u\n",&a);
getch();
}
/*Note:if the divisor bigger than dividend then dividend is the answer*/
```



```
TC
a value 10
a address 65500
```

() AND , SEPARATORS:



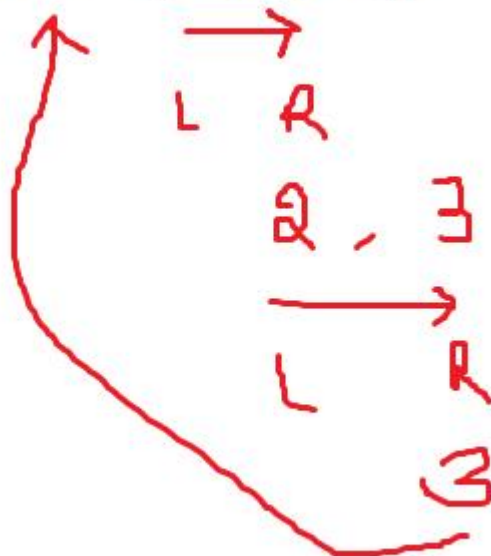
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The "Edit" menu is currently open. The main editing area has a dark blue background with yellow text. At the top of this area, a red error message reads: "Error: Declaration syntax error in function main". Below this, the following C code is visible:

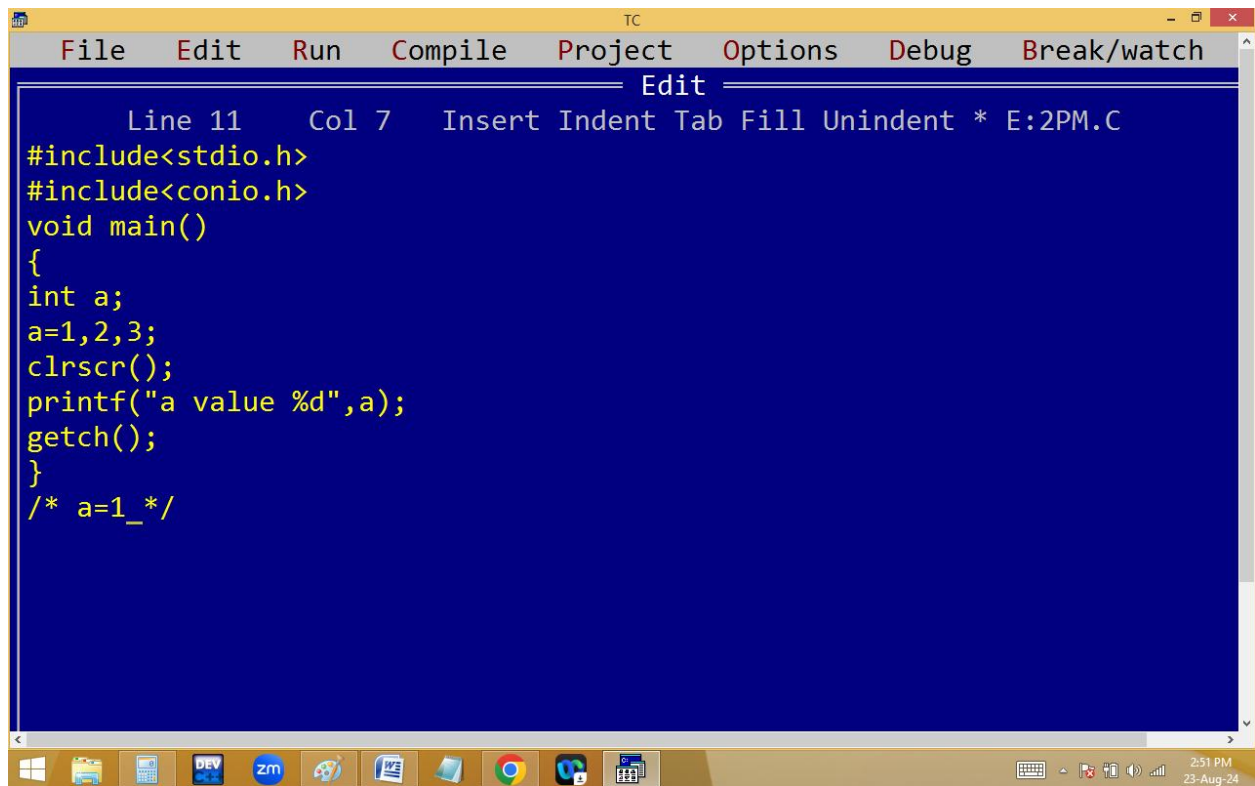
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1,2,3;
clrscr();
printf("a value %d",a);
getch();
}
/* Error */
```

The Windows taskbar is visible at the bottom, showing icons for various applications including Windows Explorer, DEV C++, Zm, and Google Chrome. The system clock in the bottom right corner indicates the time is 2:47 PM on 23-Aug-24.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 10 Col 7 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=(1,2,3);
clrscr();
printf("a value %d",a);
getch();
}
/* a=3_*/
```

int a = (1, 2, 3);





```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 11 Col 7 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=1,2,3;
clrscr();
printf("a value %d",a);
getch();
}
/* a=1_*/
```



a=1, 2, 3;

```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 6 Col 8 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=1,(2),3;
clrscr();
printf("a value %d",a);
getch();
}
/* a=1 */
```

a = 1 , (2) , 3 ;
↓
a = 1 , 2 , 3 ;

```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 6 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=1,(2,3);
clrscr();
printf("a value %d",a);
getch();
}
/* a=1 */
```

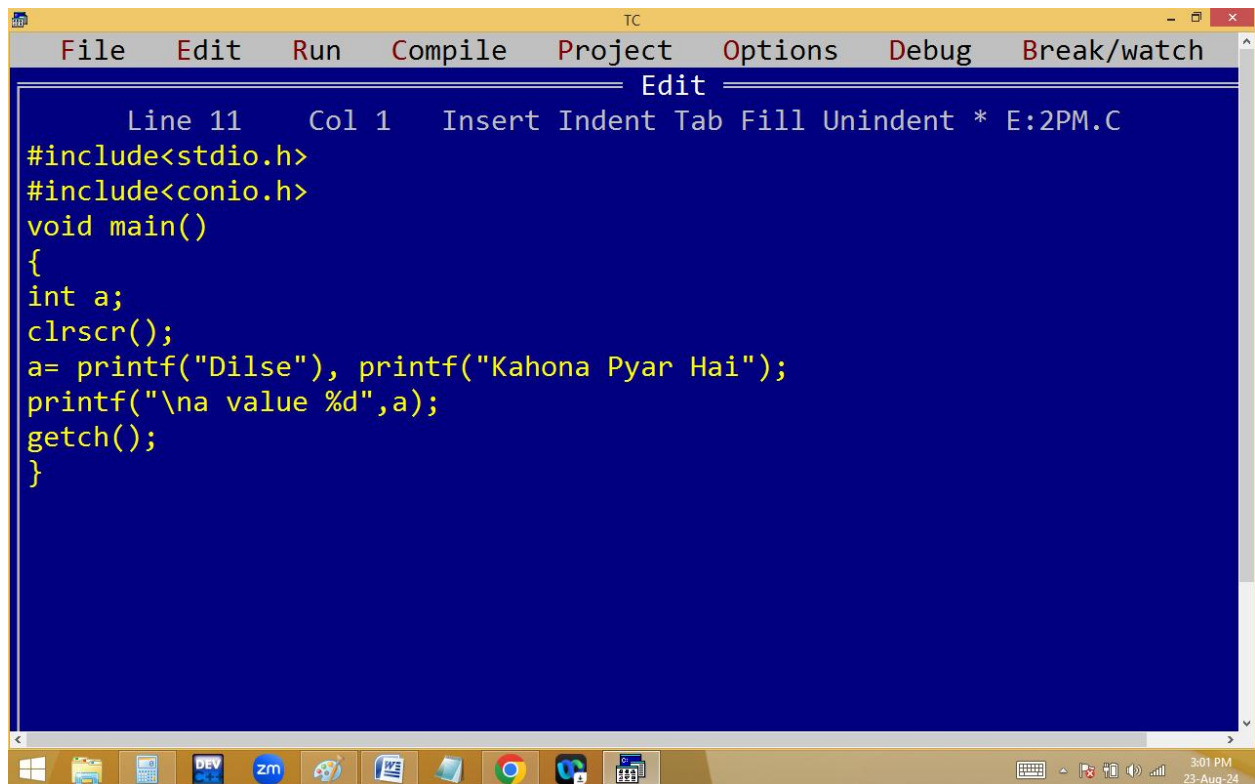
a = 1, (2, 3);


a = 1, 3

```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 11 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
a=(1,2),(3,4);
clrscr();
printf("a value %d",a);
getch();
}
/* a value 2_*/
```

a = (1,2) , (3,4);

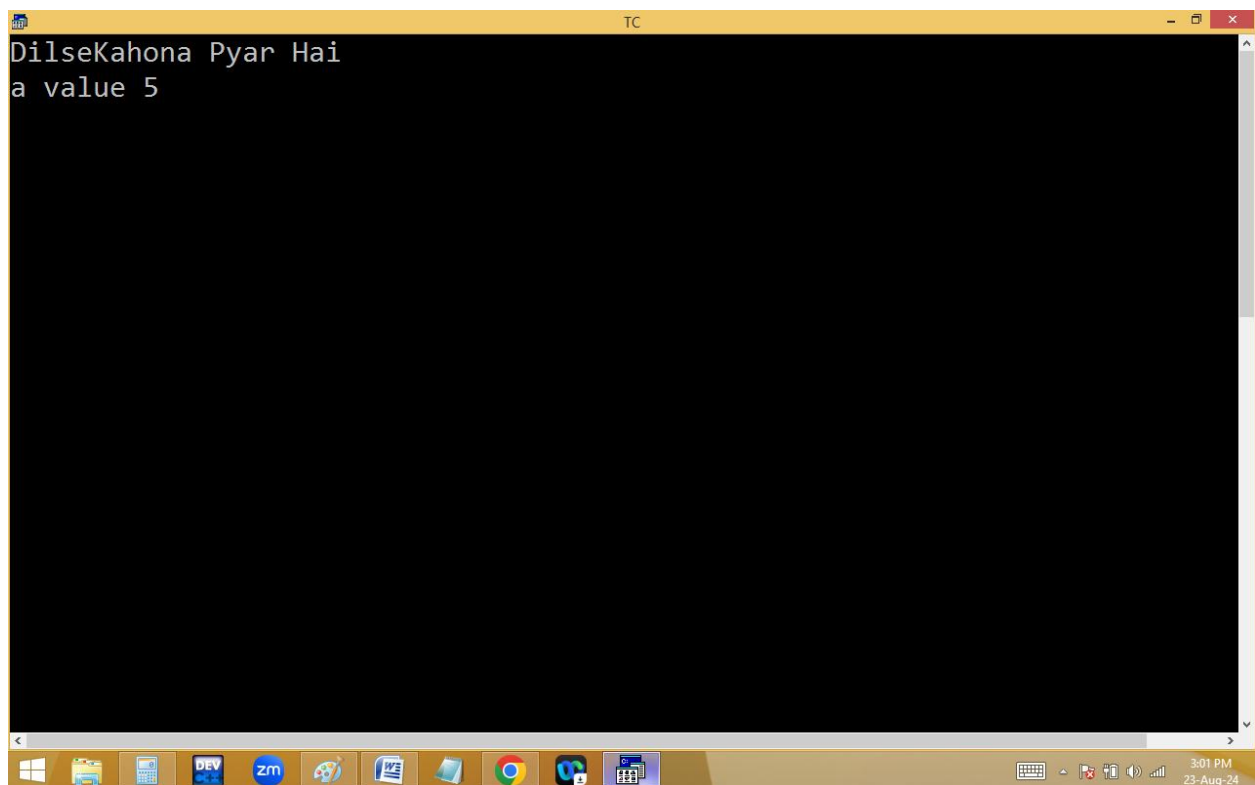
a = 2, 4



The screenshot shows the Turbo C++ IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a toolbar. The main window displays the source code for a file named E:2PM.C. The code is as follows:

```
Line 11 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
a= printf("Dilse"), printf("Kahona Pyar Hai");
printf("\na value %d",a);
getch();
}
```

The Windows taskbar at the bottom shows the Start button and several application icons, including DEV, zm, and a calendar icon. The system clock indicates 3:01 PM on 23-Aug-24.



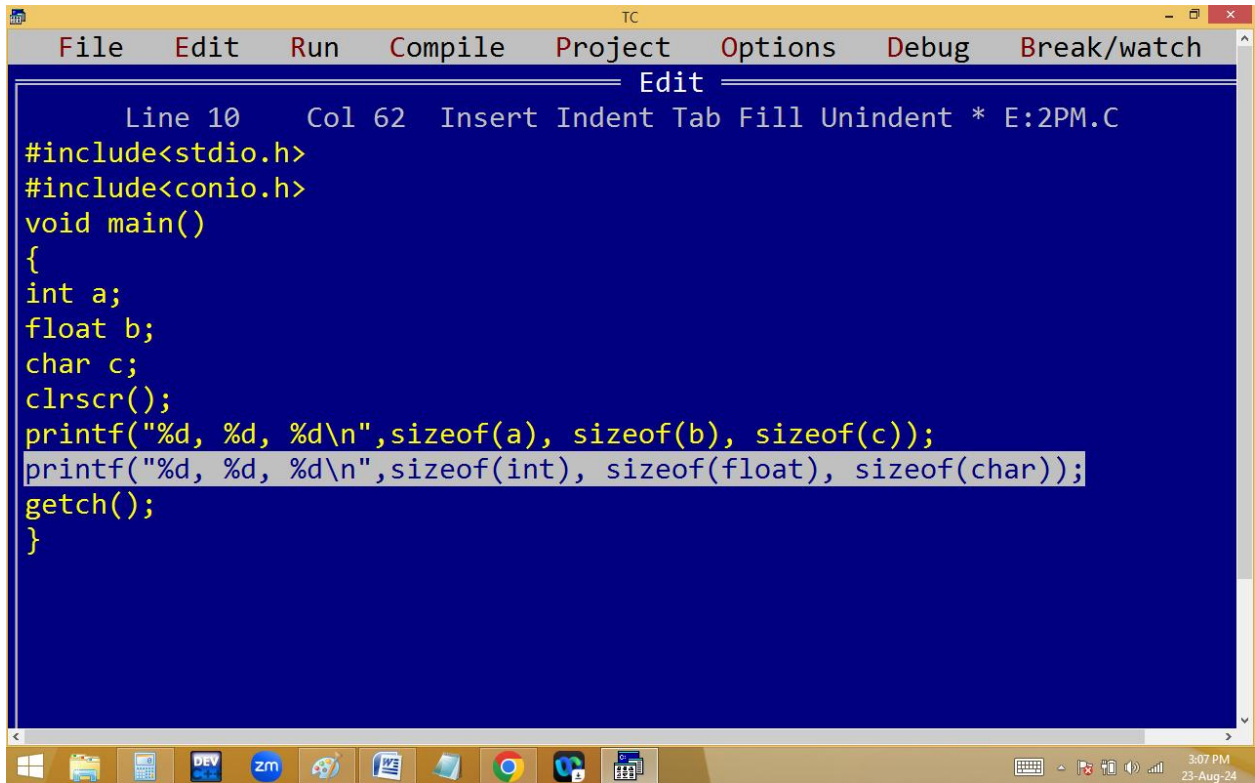
The screenshot shows the Turbo C++ IDE with the same menu bar and toolbar. The main window displays the output of the program, which is:

```
DilseKahona Pyar Hai
a value 5
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the Start button, application icons, and the system clock at 3:01 PM on 23-Aug-24.

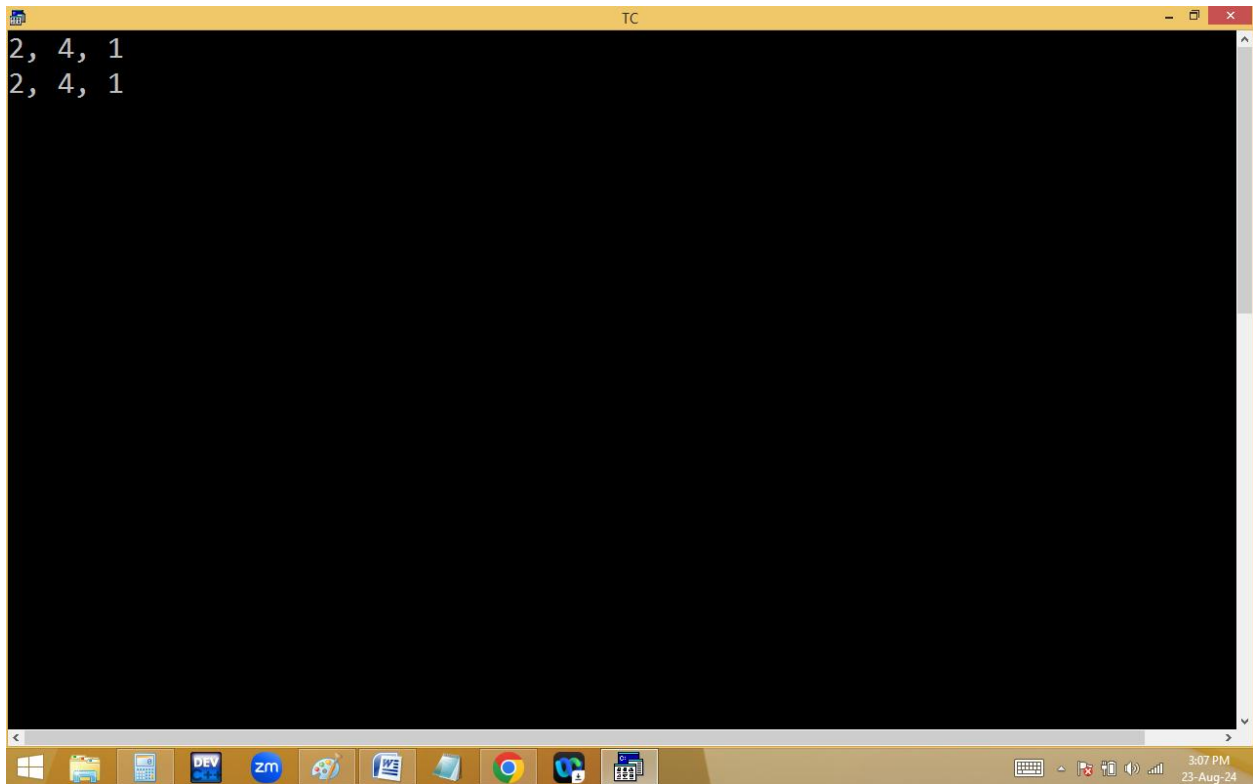
sizeof():

It returns the no of bytes taken by a variable / data type / value.



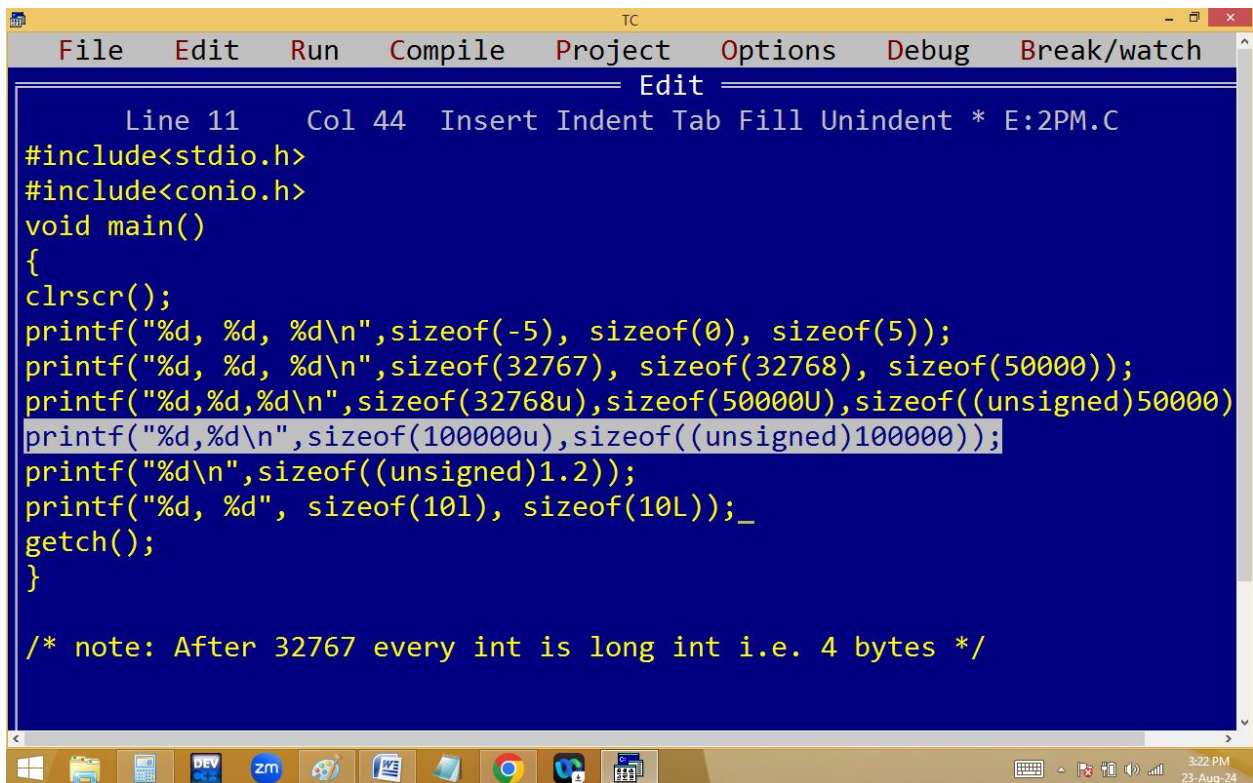
```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 10 Col 62 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
float b;
char c;
clrscr();
printf("%d, %d, %d\n",sizeof(a), sizeof(b), sizeof(c));
printf("%d, %d, %d\n",sizeof(int), sizeof(float), sizeof(char));
getch();
}
```

The screenshot shows a Turbo C++ IDE window titled 'TC'. The menu bar includes 'File', 'Edit', 'Run', 'Compile', 'Project', 'Options', 'Debug', and 'Break/watch'. The 'Edit' menu is open, showing options like 'Line 10', 'Col 62', 'Insert', 'Indent', 'Tab', 'Fill', 'Unindent', and '* E:2PM.C'. The main text area contains a C program that uses the 'sizeof' operator to determine the size of variables 'a' (int), 'b' (float), and 'c' (char), as well as the data types 'int', 'float', and 'char'. The program also includes 'clrscr()' and 'getch()' for screen clearing and user input. The Windows taskbar at the bottom shows various icons, including 'DEV', 'zm', and 'Google', along with the system clock indicating 3:07 PM on 23-Aug-24.



A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC" and standard window controls. The main area is black with white text. The output of the program is displayed on the first two lines: "2, 4, 1" followed by "2, 4, 1". The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:07 PM on 23-Aug-24.

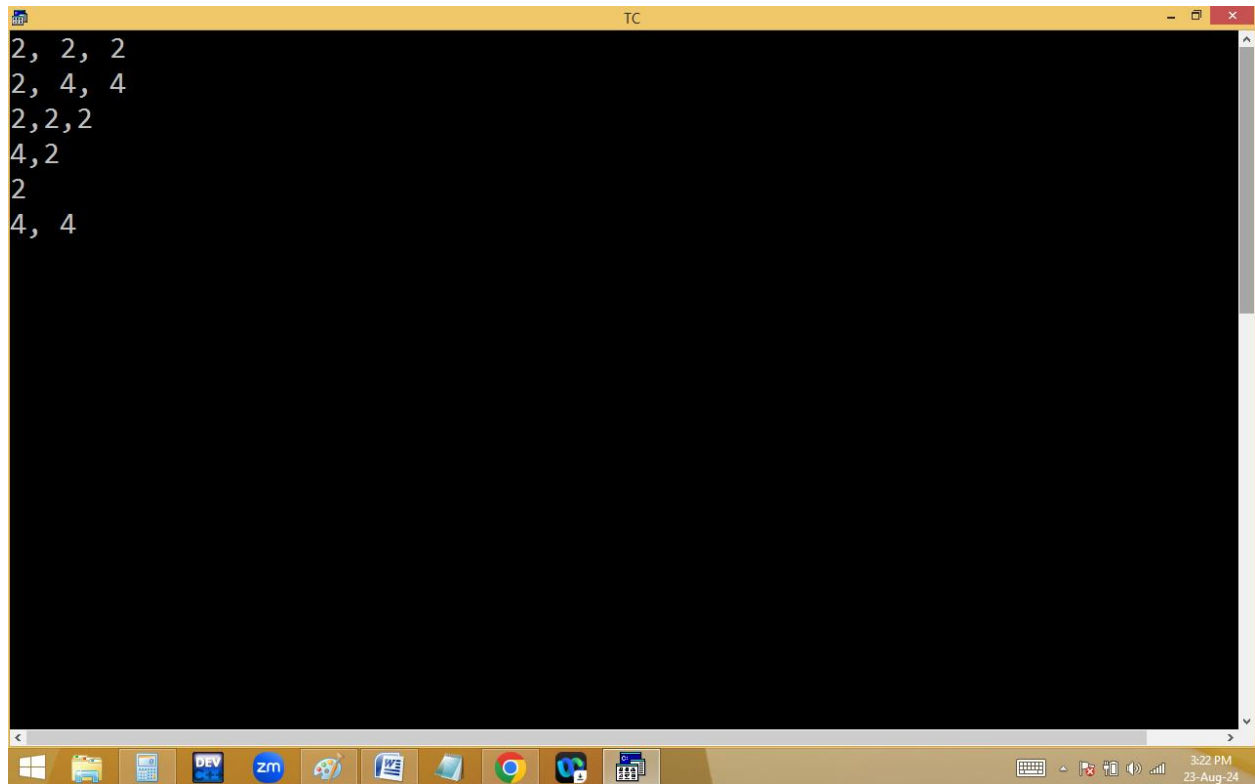
```
2, 4, 1
2, 4, 1
```



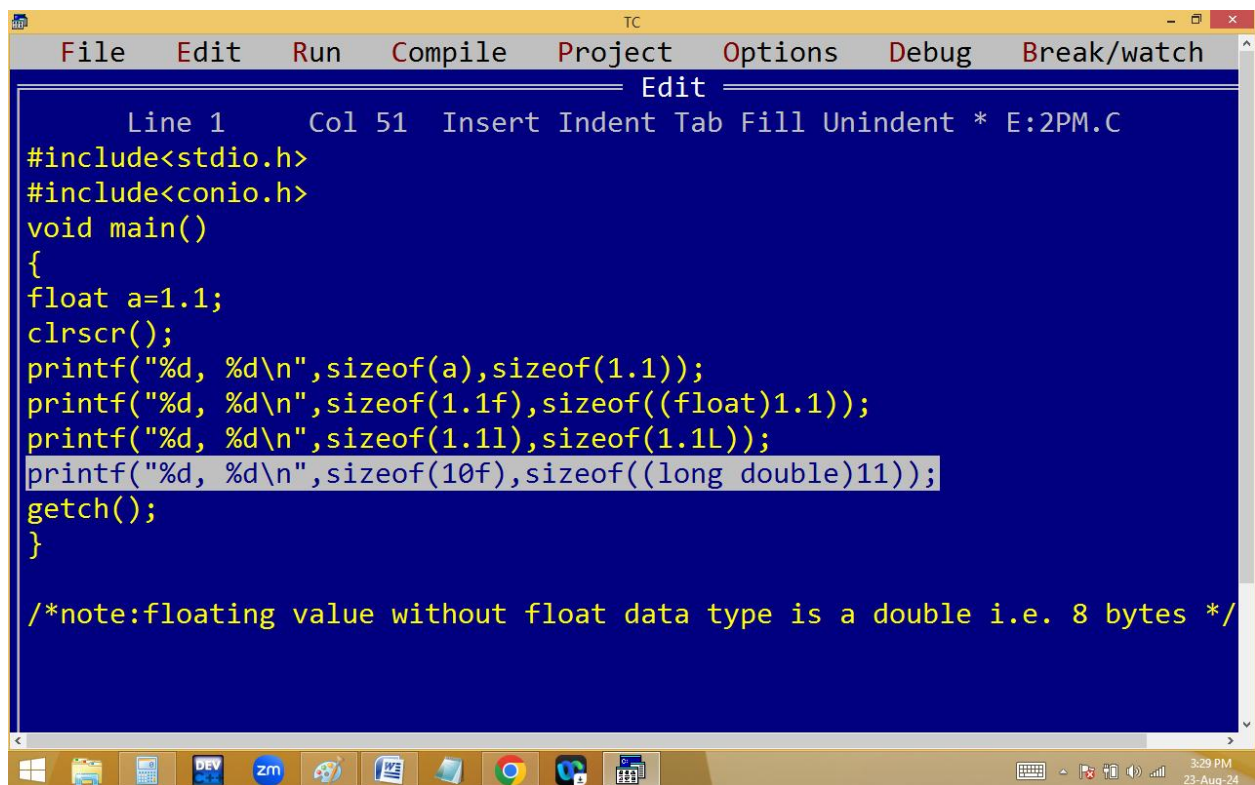
A screenshot of the Turbo C++ (TC) source code editor. The window has a yellow title bar with "TC" and standard window controls. Below the title bar is a menu bar with "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A status bar at the top of the editor shows "Line 11 Col 44 Insert Indent Tab Fill Unindent * E:2PM.C". The editor area has a blue background with yellow text. The code is a C program that includes `<stdio.h>` and `<conio.h>`, and defines a `main` function. The function uses `clrscr()` to clear the screen and several `printf` statements to print the results of `sizeof` operations. The last line of code is a comment: `/* note: After 32767 every int is long int i.e. 4 bytes */`. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:22 PM on 23-Aug-24.

```
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 11 Col 44 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d, %d, %d\n",sizeof(-5), sizeof(0), sizeof(5));
printf("%d, %d, %d\n",sizeof(32767), sizeof(32768), sizeof(50000));
printf("%d,%d,%d\n",sizeof(32768u),sizeof(50000U),sizeof((unsigned)50000));
printf("%d,%d\n",sizeof(100000u),sizeof((unsigned)100000));
printf("%d\n",sizeof((unsigned)1.2));
printf("%d, %d", sizeof(101), sizeof(10L));_
getch();
}

/* note: After 32767 every int is long int i.e. 4 bytes */
```

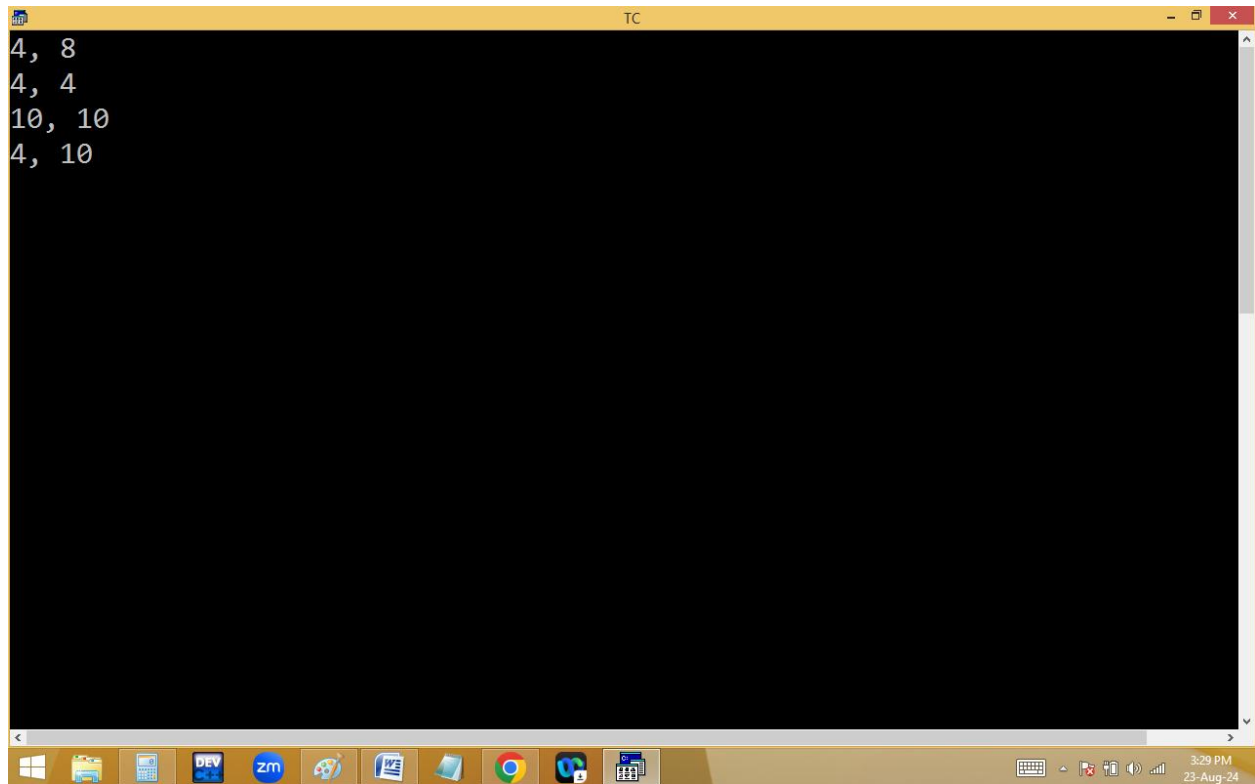


```
2, 2, 2
2, 4, 4
2,2,2
4,2
2
4, 4
```



```
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 1 Col 51 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float a=1.1;
clrscr();
printf("%d, %d\\n",sizeof(a),sizeof(1.1));
printf("%d, %d\\n",sizeof(1.1f),sizeof((float)1.1));
printf("%d, %d\\n",sizeof(1.1l),sizeof(1.1L));
printf("%d, %d\\n",sizeof(10f),sizeof((long double)11));
getch();
}

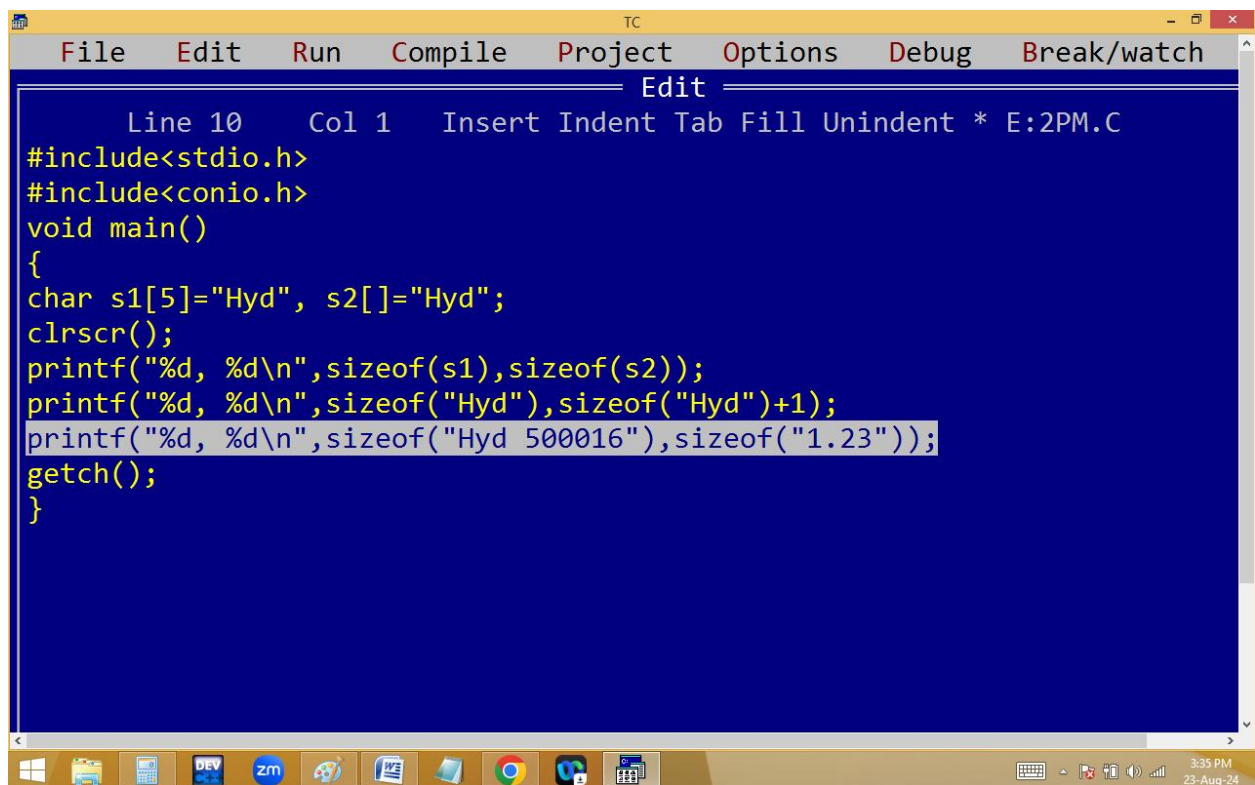
/*note:floating value without float data type is a double i.e. 8 bytes */
```



A screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" and standard window controls. The main area is black and contains the following text:

```
4, 8
4, 4
10, 10
4, 10
```

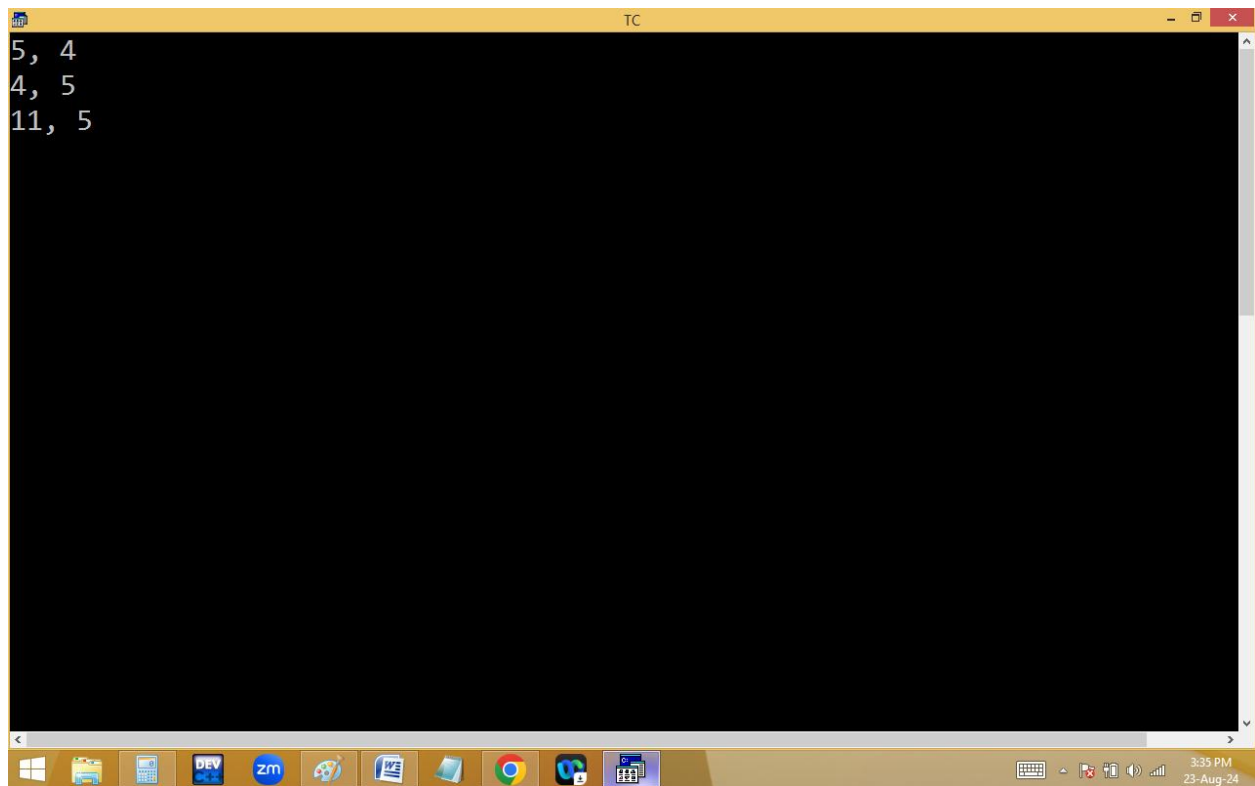
The Windows taskbar is visible at the bottom, showing icons for various applications and the system clock indicating 3:29 PM on 23-Aug-24.



A screenshot of a Turbo C++ (TC) window showing a C program source code. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, Debug, Break/watch. Below the menu bar is a toolbar with icons for various editing and execution actions. The main area is blue and contains the following C code:

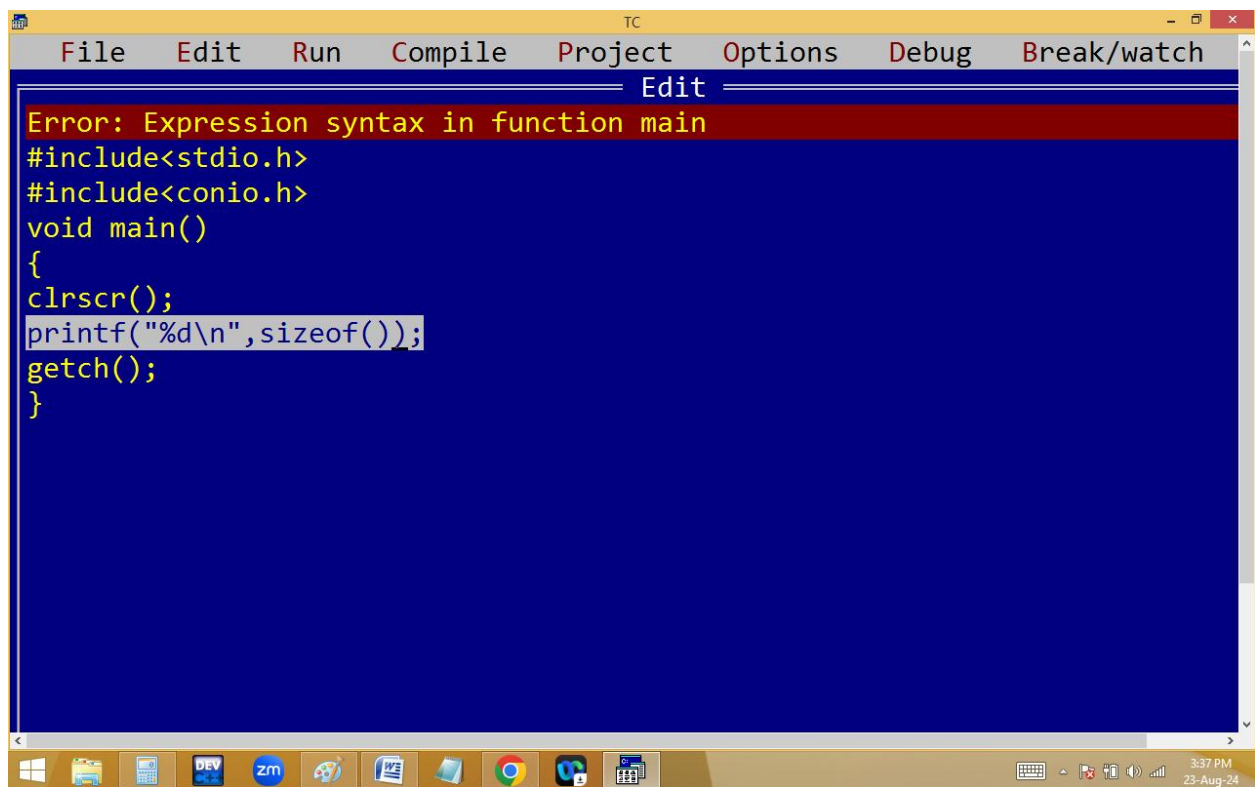
```
Line 10 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s1[5]="Hyd", s2[]="Hyd";
clrscr();
printf("%d, %d\n",sizeof(s1),sizeof(s2));
printf("%d, %d\n",sizeof("Hyd"),sizeof("Hyd")+1);
printf("%d, %d\n",sizeof("Hyd 500016"),sizeof("1.23"));
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for various applications and the system clock indicating 3:25 PM on 23-Aug-24.



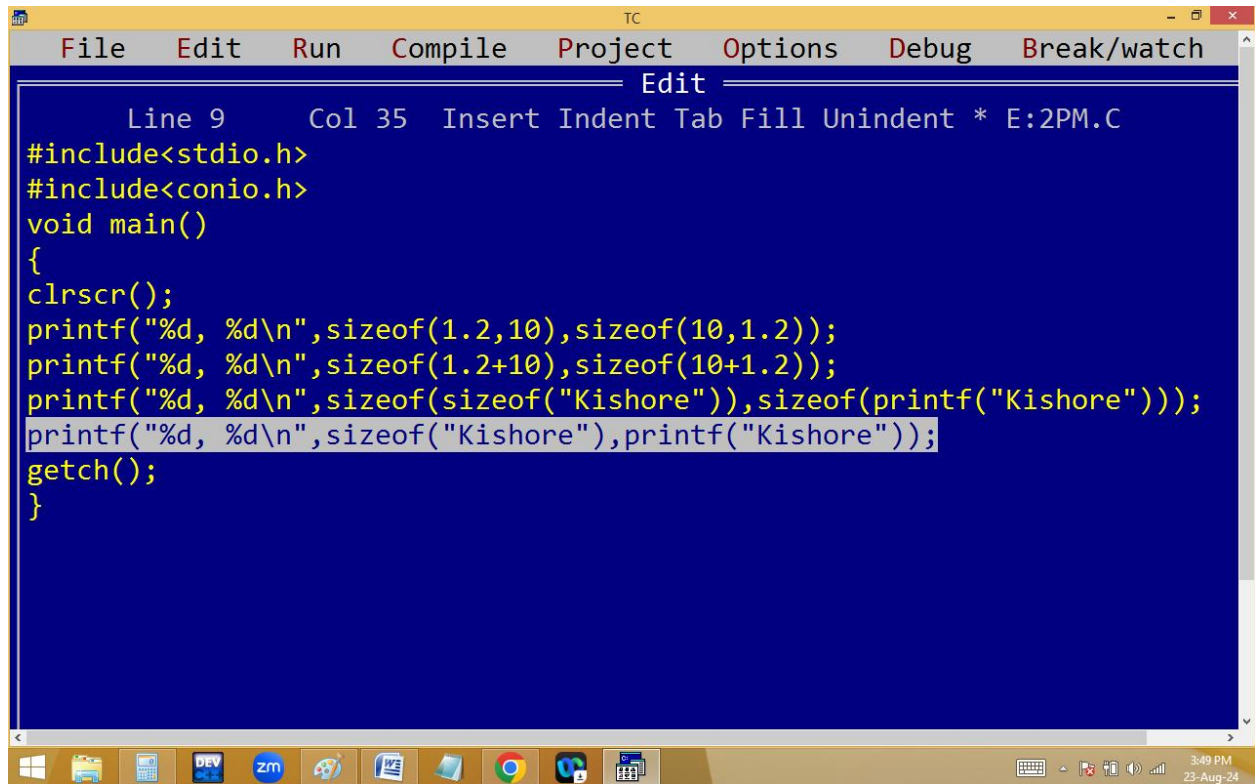
A screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" and standard window controls. The main area is black with white text. The text consists of three lines: "5, 4", "4, 5", and "11, 5". The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:35 PM on 23-Aug-24.

```
5, 4
4, 5
11, 5
```



A screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The main area has a blue background with white text. The text is a C program with a syntax error. The error message "Error: Expression syntax in function main" is displayed at the top in red. The program code is as follows: `#include<stdio.h>`, `#include<conio.h>`, `void main()`, `{`, `clrscr();`, `printf("%d\n",sizeof());`, `getch();`, `}`. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:37 PM on 23-Aug-24.

```
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof());
getch();
}
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 9 Col 35 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d, %d\\n", sizeof(1.2,10), sizeof(10,1.2));
printf("%d, %d\\n", sizeof(1.2+10), sizeof(10+1.2));
printf("%d, %d\\n", sizeof(sizeof("Kishore")), sizeof(printf("Kishore")));
printf("%d, %d\\n", sizeof("Kishore"), printf("Kishore"));
getch();
}
```



```
TC
2, 8
8, 8
2, 2
Kishore8, 7
_
```

